

**KENDRIYA VIDYALAYA SANGATHAN
REGIONAL OFFICE PATNA
FIRST PRE-BOARD EXAMINATION 2024-25
ARTIFICIAL INTELLIGENCE (SUBJECT CODE - 417)
CLASS X (SESSION 2024-2025)
(QUESTION PAPER)**

Max. Time: 2 Hours

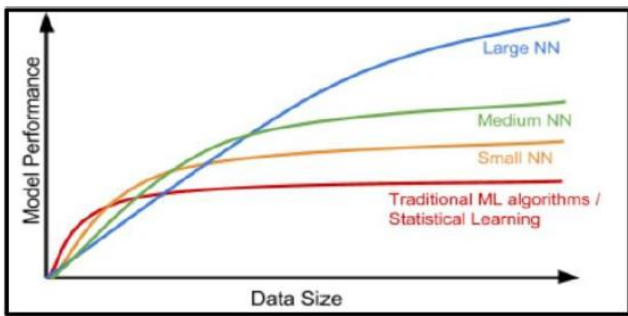
Max. Marks: 50

General Instructions:

- 1. Please read the instructions carefully.**
- 2. This Question Paper consists of 21 questions in two sections: Section A & Section B.**
- 3. Section A has Objective type questions whereas Section B contains Subjective type questions.**
- 4. Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.**
- 5. All questions of a particular section must be attempted in the correct order.**
- 6. SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
 - i. This section has 05 questions.**
 - ii. Marks allotted are mentioned against each question/part.**
 - iii. There is no negative marking.**
 - iv. Do as per the instructions given.**
- 7. SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
 - i. This section has 16 questions.**
 - ii. A candidate has to do 10 questions.**
 - iii. Do as per the instructions given.**
 - iv. Marks allotted are mentioned against each question/part.**

SECTION A: OBJECTIVE TYPE QUESTIONS

Q. 1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i.	<i>“Suresh participated in a 100m race and won a prize. This motivated him to go for practice every morning”.</i> Which type of motivation is clearly visible in the given statement?	1
ii.	HOME key on keyboard is classified under (a)Punctuation Keys (b)Navigation Keys (c)Command Keys (d)Control Keys	1
iii.	Assertion (A): A doctor who works for a hospital is a self-employed person Reason(R): A self-employed person running a business to satisfy a need by trying new ideas is an entrepreneur. (a)Both A and R are correct and R is the correct explanation of A (b)Both A and R are correct but R is NOT the correct explanation of A (c)A is correct but R is not correct (d)A is not correct but R is correct	1
iv.	Which of the following is not a step to manage emotional intelligence? (a)Generate emotions (b)Understand your emotions (c)Rationalize (d)Practice	1
v.	 The above given visual communication refers to (a) Under CCTV surveillance (b) Radiation/biohazard warning (c) No mobile phone (d) No entry	1
vi.	“Sumit is using organic and natural fertilizers, such as cow dung in growing crops”. This is an example of (a) Chemical Farming (b) Organic Farming (c) Sustainable Process (d) Both (b) and (c)	1

Q. 2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Assertion (A) – An Automatic Washing Machine is an example of AI. Reason(R) – When a machine displays attributes like reasoning, learning and solving problems based on the situation or environment around it, the machine is said to be based on Artificial Intelligence. (a) Both A and R are correct and R is the correct explanation of A (b) Both A and R are correct but R is not the correct explanation of A (c) A is correct but R is not correct (d) A is not correct but R is correct.	1
ii.	“Siri is an example of _____. ” Fill in the blanks with appropriate option. (a) Social Media (b) Space Exploration (c) Chatbot (d) All of them	1
iii.	Statement1: Neural Network systems are modelled on the human brain and nervous system. Statement2: Hidden layers of neural network perform all the processing. (a) Both Statement1 and Statement2 are correct (b) Both Statement1 and Statement2 are incorrect (c) Statement1 is correct but Statement2 is incorrect (d) Statement2 is correct but Statement1 is incorrect	1
iv.	Observe the given graph and fill in the blank:  _____ stop improving after a certain saturation point.	1
v.	_____ is a language which is helpful in handling structured data. (a) DBMS (b) SQL (c) Spreadsheet (d) CSV	1

vi.	Ahana is given a long text passage. She extracts all the unique words from this passage and also the number of times they occur in it. This feature of Natural Language Processing is known as (a) Stop words (b) Tokenization (c) Stemming (d) Bag of words	1
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Q. 3	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Read the examples given below- i. Automatic Toy Car ii. Face lock in smartphones iii. Air Conditioner iv. Email filters Choose the options that are AI (a) i and ii (b) iii and i (c) ii and iv (d) i, iii and iv	1
ii.	_____ convert real world phenomenon like temperature/force to voltage or current signals that can be used as inputs.	1
iii.	Naman is new to the field of Data Science and wants to perform data acquisition. Which of the following sources he could use? (a) Surveys (b) Web Scrapping (c) Cameras (d) All of them	1
iv.	Atul is using QR code scanner to make payment on a shop. He is basically making use of the technique of (a) Computer Vision (b) Stemming (c) Lemmatization (d) Resolution	1
v.	Identify the given NLP application It helps us to convert one language text (say English) into another (say Hindi). (a) Text summarization (b) Sentiment Analysis (c) Machine Translation (d) Parts of speech tagging	1

vi.	F1 score is harmonic mean of (a) Precision and False Positive Rate (b) Precision and Recall (c) Recall and Accuracy (d) Precision and Accuracy	1
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Q. 4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Anita has recently started learning about AI. She has forgotten the name of the domain of AI, which is multidisciplinary field that combines mathematics, statistics and computer science. Help her with the name of domain.	1
ii.	Statment1: If we want high precision, we need to reduce False Positives (FP). Statement2: If we want high recall, we need to reduce False Negatives (FN). (a) Both Statement1 and Statement2 are correct (b) Both Statement1 and Statement2 are incorrect (c) Statement1 is correct but Statement2 is incorrect (d) Statement2 is correct but Statement1 is incorrect	1
iii.	Raman has to design a deep learning-based model, which will translate English sentences into Hindi. He collects 10000 instances of English-Hindi pairs to design his model. Out of 10000 instances, he uses 9000 instances to build the model and remaining 1000 instances to check performance of his model. Which of the following statements is incorrect? (a) Size of training data is 9000. (b) Size of test data is 1000. (c) Size of training data is larger than test data. (d) Size of training data is 10000.	1
iv.	The closeness of a captured image to the original image is decided by the number of _____.	1
v.	Which model will always have an optimum accuracy? (a) Underfitted (b) Overfitted (c) Good fit (d) Both (a) and (b)	1
vi.	A machine was given a banana and asked whether it is banana. The machine correctly predicted it as “Yes”. This is a case of (a) True Positive (b) False Positive (c) True Negative (d) False Negative	1

Q. 5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Ahana has developed a model, which classifies social media comments into positive and negative. Identify the most appropriate domain of AI in the given scenario (a)Computer Vision (b)Data Science (c)Natural Language Processing (d)None of these	1
ii.	Write Accuracy in terms of True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN).	1
iii.	The full form of ppi in terms of resolution is (a) Pixels Per Inch (b) Pixels Per Image (c)Picture Per Inch (d)Picture Per Image	1
iv.	The process which always reduces a larger word into a meaningful smaller word is known as (a) Stemming (b)Lemmatization (c)Both (a) and (b) (d)Stop word	1
v.	Which process divides a large corpus into sentences. (a)Stemming (b)Tokenization (c)Text normalization (d)Sentence Segmentation	1
vi.	The best method to measure as to how successful a model is (a)F1 Score (b)Reality (c)Prediction (d)Precision	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills

(2 x 3 = 6 marks) Answer each question in 20 – 30 words.

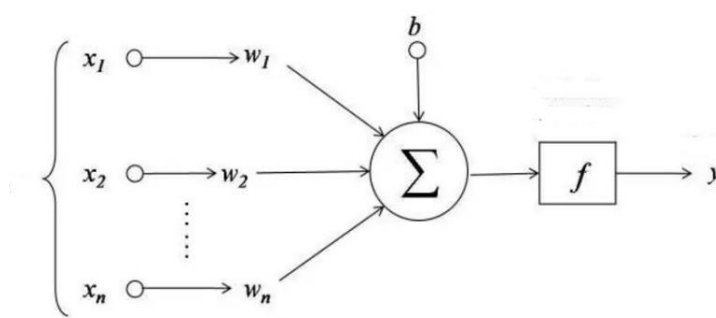
Q. 6	Give any two examples of non-verbal communication.	2
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Q. 7	Mention external and internal factors which cause stress.	2
Q. 8	Sumit is experiencing slow working of his computer system, as virus has entered his computer system. Mention four possible causes due to which virus would have entered the system.	2
Q. 9	Aman wants to become a successful entrepreneur in future. Name any four qualities he needs to possess in order to become a successful entrepreneur?	2
Q. 10	What measures an individual or community should adopt for creation of a sustainable city?	2

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

Q. 11	Several e-commerce sites, such as Flipkart, Amazon, Snapdeal, are using AI technology for selling products. Explain how?	2
Q. 12	Consider the undermentioned situations and mention the type of learning that one should use in each of the given situation. (a) Harman is given a set of labelled emails and asked to classify a new email into spam or not-spam. (b) ABC company has customer data and wants to group the customers into separate groups based on their similarities. (c) A company wants to find out people who by product X also tend to buy product Y. (d) A student is given a set of images, labelled as cat, dog or rat, and asked to classify a new image into either of these categories.	2
Q. 13	Anamika is new to the field of AI and wants to know about the broad domains of AI. Name the three broad domains of AI and briefly explain any one of them.	2
Q. 14	Ishita is given an image, which can be mapped to a grid of size 400 by 400. Find the resolution of this image.	2
Q. 15	Define stemming. What is the stem word and lemmatized word for “dried”?	2
Q. 16	Draw the confusion matrix for the following data • the number of false positive = 18 • the number of false negative = 72 • the number of true positive = 26 • the number of true negative = 19	2

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

Q. 17	Briefly explain any two types of intelligence. How is human intelligence different from machine intelligence?	4													
Q. 18	Sumit wants to learn how to scope the problem for an AI Project (a) Briefly explain Who Block and What Block in 4W's problem canvas. (b) Why do we need AI project cycle?	4													
Q. 19	Consider the below given Artificial Neural Network diagram.  (a)Write expression for weighted sum of inputs. (b)Write expression for the final output, y.	4													
Q. 20	Ahana is interested in application areas of Natural Language Processing, specifically chatbots. Help her know the same by answering the following questions (a) Define chatbot. (b)Which was the first chatbot? (c)Mention two points of differences between Script bot and Smart bot.	4													
Q. 21	Consider the below given Confusion matrix and answer the questions which follow: <table border="1" data-bbox="229 1700 1339 1912"><thead><tr><th colspan="2" rowspan="2">Confusion Matrix</th><th colspan="2">Reality</th></tr><tr><th>Yes</th><th>No</th></tr></thead><tbody><tr><th rowspan="2">Prediction</th><th>Yes</th><td>20</td><td>5</td></tr><tr><th>No</th><td>10</td><td>40</td></tr></tbody></table> (i) Identify the total number of TP, TN, FP and FN. (ii) Calculate precision and recall.	Confusion Matrix		Reality		Yes	No	Prediction	Yes	20	5	No	10	40	4
Confusion Matrix				Reality											
		Yes	No												
Prediction	Yes	20	5												
	No	10	40												